

Rounded polygons: geometry, intersections, overlap, and forces

A *rounded polygon* is a backbone polygon whose corners are radius- ρ circular arcs. We use the 2D Levi-Civita symbol $\varepsilon_{\alpha\beta}$ ($\varepsilon_{xy} = -\varepsilon_{yx} = 1$), write $\vec{a} \otimes \vec{b} \equiv \varepsilon_{\alpha\beta} a^\alpha b^\beta$ for the 2D cross, and sum repeated Greek indices. Vertices are $\vec{v}_k$; $z(k), z'(k)$ are the next / previous vertex CCW. Inter-polygon position differences use the periodic minimum image.

1 Rounded-polygon geometry

At corner $\vec{v}_k$ the two incident edge units are

$$\hat{e}_k = \frac{\vec{v}_{z(k)} - \vec{v}_k}{|\vec{e}_k|} \quad (\text{outgoing}), \quad \hat{e}_{z'(k)} = \frac{\vec{v}_k - \vec{v}_{z'(k)}}{|\vec{e}_{z'(k)}|} \quad (\text{incoming}), \quad (1.1)$$

the wedge angle is $\theta_k = \arccos(-\hat{e}_{z'(k)} \cdot \hat{e}_k) \in [0, \pi]$, and the corner is rounded by a circle of radius ρ tangent to both edges. The kiss offset, kiss points, bisector, and center are

$$t_k = \frac{\rho}{\tan(\theta_k/2)}, \quad \vec{a}_k^+ = \vec{v}_k + t_k \hat{e}_k, \quad \vec{a}_k^- = \vec{v}_k - t_k \hat{e}_{z'(k)}, \quad \vec{z}_k = \vec{v}_k + \frac{\rho}{\sin(\theta_k/2)} \hat{b}_k, \quad (1.2)$$

with bisector $\hat{b}_k = (\hat{e}_k - \hat{e}_{z'(k)})/|\hat{e}_k - \hat{e}_{z'(k)}|$ and $|\hat{e}_k - \hat{e}_{z'(k)}| = 2 \cos(\theta_k/2)$. The arc C_k sweeps $\psi_k = \pi - \theta_k$, and $\hat{b}_k$ points from $\vec{v}_k$ toward $\vec{z}_k$ (into the corner when convex, outward when reflex). The rounding is feasible iff each edge fits its two offsets, $t_k + t_{z(k)} \leq |\vec{e}_k|$.

Arc-validity (AVT). The intersection formulas (§3) place candidate points on the *full* circle $(\vec{z}_k, \rho_k)$; the arc is only the wedge symmetric about $-\hat{b}_k$ spanning $\pm\psi_k/2$. A circle point $\vec{X}$ lies on C_k iff

$$\frac{\vec{X} - \vec{z}_k}{\rho_k} \cdot \hat{b}_k \leq -\cos(\psi_k/2), \quad (1.3)$$

which holds for convex and reflex corners alike (both arcs bulge toward $\vec{v}_k$).

2 Rounded area and perimeter

Number the $2n$ boundary features CCW with arcs C_k and edges E_k alternating: the edge runs $\vec{a}_k^+ \rightarrow \vec{a}_{z(k)}^-$ and the arc C_k runs $\vec{a}_k^- \rightarrow \vec{a}_k^+$.

Perimeter. Straight runs plus arc lengths,

$$P = \sum_k (|\vec{e}_k| - t_k - t_{z(k)}) + \sum_k \rho_k \psi_k. \quad (2.1)$$

Area. Read the rounded area as the polygon through the kiss points (every corner cut by its chord $\vec{a}_k^- \rightarrow \vec{a}_k^+$) *plus* a circular segment per corner:

$$A = A_{\text{kiss polygon}} + \sum_k s_k \frac{1}{2} \rho_k^2 (\psi_k - \sin \psi_k), \quad s_k = \begin{cases} +1 & \text{convex} \\ -1 & \text{reflex,} \end{cases} \quad (2.2)$$

the segment added at convex corners (the arc bulges past the chord) and subtracted at reflex ones. This “polygon + segments” reading is the same one the overlap area uses (§4).

3 Boundary intersections and outersections

Two rounded polygons A, B meet at four feature combinations: edge–edge (**ee**), edge–arc (**ea**), arc–edge (**ae**), arc–arc (**aa**).

Periodic pair shift. Positions live in the unit square $[0, 1]^2$ with periodic walls. We work in A ’s frame and translate B to its nearest image: with representative vertices $\vec{r}_i \in A$, $\vec{r}_j \in B$, the *pair shift* is $\vec{T}_{ij} = \text{wrap}(\vec{r}_j - \vec{r}_i) - (\vec{r}_j - \vec{r}_i)$, applied once to every point feature of B ($\vec{z}_j \rightarrow \vec{z}_j + \vec{T}_{ij}$, $\vec{a}_j^\pm \rightarrow \vec{a}_j^\pm + \vec{T}_{ij}$); we keep only this single image. (In code, **shift**.)

Features. The edge and arc are parametrized

$$\vec{P}_k(s) = \vec{a}_k^+ + s \vec{u}_k, \quad \vec{u}_k = \vec{a}_{z(k)}^- - \vec{a}_k^+, \quad s \in [0, 1]; \quad \vec{Q}_k(\phi) = \vec{z}_k + \rho_k(\cos \phi, \sin \phi), \quad (3.1)$$

with $\phi_k^\pm = \text{atan2}(\vec{a}_k^\pm - \vec{z}_k)$, $\psi_k = \phi_k^+ - \phi_k^-$.

Boundary coordinate and outersection. Number the $2n$ features $0, 1, \dots, 2n - 1$; an intersection on feature f of ∂A gets

$$\sigma_A = f + \begin{cases} \lambda & f \text{ even (arc)} \\ s & f \text{ odd (edge)}, \end{cases} \quad \lambda = \frac{\phi - \phi_k^-}{\psi_k}. \quad (3.2)$$

Sorting a pair’s intersections by σ_A gives the order met walking ∂A CCW; the *outersection* of an intersection is the next one in that cyclic order.

ee. For edges k, ℓ with $\vec{w}_{k\ell} = \vec{a}_\ell^+ - \vec{a}_k^+$, the two edge fractions ($\vec{P}_k(s_k) = \vec{P}_\ell(s_\ell)$) are

$$s_k = \frac{\varepsilon_{\alpha\beta} w_{k\ell}^\alpha u_\ell^\beta}{\varepsilon_{\alpha\beta} u_k^\alpha u_\ell^\beta}, \quad s_\ell = \frac{\varepsilon_{\alpha\beta} w_{k\ell}^\alpha u_k^\beta}{\varepsilon_{\alpha\beta} u_k^\alpha u_\ell^\beta}, \quad (3.3)$$

valid iff the denominator $\neq 0$ and $s_k, s_\ell \in [0, 1]$.

ea / ae. Edge E_k meets the full circle of arc ℓ ; with $\vec{w}_{k\ell} = \vec{a}_k^+ - \vec{z}_\ell$,

$$S_\pm = \frac{-(\vec{w}_{k\ell} \cdot \vec{u}_k) \pm \sqrt{D_{k\ell}}}{|\vec{u}_k|^2}, \quad D_{k\ell} = (\vec{w}_{k\ell} \cdot \vec{u}_k)^2 + |\vec{u}_k|^2 (\rho_\ell^2 - |\vec{w}_{k\ell}|^2). \quad (3.4)$$

Keep each root in $[0, 1]$ whose point passes the AVT for arc ℓ (an edge can hit one arc twice); **ae** swaps the roles.

aa. A point on both circles lies on the radical line; with $\vec{n}_{k\ell} = \vec{z}_\ell - \vec{z}_k$, $d = |\vec{n}_{k\ell}|$, $\hat{n} = \vec{n}_{k\ell}/d$,

$$a = \frac{d^2 + \rho_k^2 - \rho_\ell^2}{2d}, \quad h = \sqrt{\rho_k^2 - a^2}, \quad \vec{X}_\pm = \vec{z}_k + a \hat{n} \pm h \hat{n}^\perp, \quad (3.5)$$

real iff $|\rho_k - \rho_\ell| \leq d \leq \rho_k + \rho_\ell$; keep each $\vec{X}_\pm$ passing the AVT for both arcs.

4 Overlap area

The overlap region $A \cap B$ has area $\frac{1}{2} \varepsilon_{\alpha\beta} \oint_{\partial(A \cap B)} X^\alpha dX^\beta$, accumulated feature by feature relative to a reference $\vec{R}$ = the first backbone vertex of A (the total is $\vec{R}$ -invariant; $\vec{R}$ just keeps the arithmetic local). Each feature piece $\vec{P} \rightarrow \vec{Q}$ contributes the chord triangle from $\vec{R}$,

$$h_e(\vec{P}, \vec{Q}) = \frac{1}{2} \varepsilon_{\alpha\beta} (P - R)^\alpha (Q - R)^\beta, \quad (4.1)$$

and an *arc* piece adds its circular segment

$$h_a(\vec{P}, \vec{Q}) = \frac{1}{2} \rho^2 (\theta - \sin \theta), \quad \theta = \text{atan2}(\varepsilon_{\alpha\beta} (P - z)^\alpha (Q - z)^\beta, (P - z) \cdot (Q - z)), \quad (4.2)$$

the signed central angle of the sub-arc. So each feature contributes

$$h(\vec{P}, \vec{Q}) = \begin{cases} h_e(\vec{P}, \vec{Q}) & \text{edge} \\ h_e(\vec{P}, \vec{Q}) + h_a(\vec{P}, \vec{Q}) & \text{arc:} \end{cases} \quad (4.3)$$

the boundary's chord-polygon (all the h_e) plus a segment on every arc – the same “polygon + segments” picture as §2.

Traversal. The overlap boundary is the kept runs of ∂A inside B and ∂B inside A , each from an intersection to its outersection. Label an intersection pair $P = \{i, j, k, l\}$: feature i (on A) meets j (on B), and following i CCW reaches feature l , which meets k . With $\sigma^- \equiv \lfloor \sigma \rfloor$ the feature index and $\vec{q}$ the intersection / kiss points,

$$U = \sum_P \left\{ \delta_{i=-l-} h(\vec{q}_i, \vec{q}_l) + (1 - \delta_{i=-l-}) \left[h(\vec{q}_i, \vec{q}_{z(i-)}) + \sum_{m=z(i-)}^{z'(l-)} h(\vec{q}_m, \vec{q}_{z(m)}) + h(\vec{q}_{l-}, \vec{q}_l) \right] \right\}. \quad (4.4)$$

A run on one feature gives $h(\vec{q}_i, \vec{q}_l)$; otherwise the partial start feature, the whole interior features, and the partial end feature.

Parallel split. Peel off the interior sum, $U = U_{\text{out}} + U_{\text{in}}$ with U_{out} per intersection pair (the two partial end features) and $U_{\text{in}} = \sum_m \sum_P \delta_{m \in \{z(i-), \dots, z'(l-)\}} h(\vec{q}_m, \vec{q}_{z(m)})$ per interior feature m . The walk and the split agree to machine precision and match Monte Carlo to $\sim 0.1\%$ (convex) / $\sim 0.5\%$ (non-convex).

5 Overlap forces

The overlap energy is $K_o U$ (§8), so the force is $\vec{F} = -K_o \partial U / \partial \vec{v}$, and $U = \sum_{\text{pieces}} h$. Differentiating a piece, with its endpoints $\vec{P}, \vec{Q}$ moving and (on an arc) its center $\vec{z}$,

$$\frac{\partial h}{\partial v_k^\alpha} = \frac{\partial h}{\partial P^\beta} \frac{\partial P^\beta}{\partial v_k^\alpha} + \frac{\partial h}{\partial Q^\beta} \frac{\partial Q^\beta}{\partial v_k^\alpha} + \mathbf{1}_{\text{arc}} \frac{\partial h_a}{\partial z^\beta} \frac{\partial z^\beta}{\partial v_k^\alpha}, \quad (5.1)$$

where $\partial h / \partial P = \partial h_e / \partial P$ on an edge and $\partial h_e / \partial P + \partial h_a / \partial P$ on an arc (likewise Q). $\vec{R}$ -invariance lets us drop $d\vec{R}$.

Edge piece. From $h_e = \frac{1}{2}\varepsilon_{\alpha\beta}(P-R)^\alpha(Q-R)^\beta$,

$$\frac{\partial h_e}{\partial P^\gamma} = \frac{1}{2}\varepsilon_{\gamma\beta}(Q^\beta - R^\beta), \quad \frac{\partial h_e}{\partial Q^\gamma} = -\frac{1}{2}\varepsilon_{\gamma\beta}(P^\beta - R^\beta). \quad (5.2)$$

Arc segment. $\partial h_a/\partial(\cdot) = \frac{1}{2}\rho^2(1 - \cos\theta)\partial\theta/\partial(\cdot)$, and since $\vec{P}, \vec{Q}$ are on the circle, $\cos\theta = (P-z)\cdot(Q-z)/\rho^2$. Implicit differentiation of $\tan\theta$ with the Schouten identity $A^\alpha\varepsilon_{\gamma\beta} - A^\gamma\varepsilon_{\alpha\beta} = A^\beta\varepsilon_{\gamma\alpha}$ gives

$$\frac{\partial\theta}{\partial P^\gamma} = \frac{1}{\rho^2}(P^\alpha - z^\alpha)\varepsilon_{\gamma\alpha}, \quad (5.3)$$

$$\frac{\partial\theta}{\partial Q^\gamma} = -\frac{1}{\rho^2}(Q^\alpha - z^\alpha)\varepsilon_{\gamma\alpha}, \quad (5.4)$$

$$\frac{\partial h_a}{\partial P^\gamma} = \frac{1}{2}(1 - \cos\theta)(P^\alpha - z^\alpha)\varepsilon_{\gamma\alpha}, \quad \frac{\partial h_a}{\partial Q^\gamma} = -\frac{1}{2}(1 - \cos\theta)(Q^\alpha - z^\alpha)\varepsilon_{\gamma\alpha}. \quad (5.5)$$

For the center, θ depends only on $\vec{P} - \vec{z}, \vec{Q} - \vec{z}$, so $\partial\theta/\partial\vec{P} + \partial\theta/\partial\vec{Q} + \partial\theta/\partial\vec{z} = 0$ and

$$\frac{\partial h_a}{\partial z^\gamma} = -\frac{1}{2}(1 - \cos\theta)\varepsilon_{\gamma\alpha}(P^\alpha - Q^\alpha). \quad (5.6)$$

6 Derivatives of features

The endpoints $\vec{P}, \vec{Q}$ are kiss points $\vec{a}^\pm$ or arc centers $\vec{z}$ (the “easy” pieces) or intersection points $\vec{q}$. We compute the kiss-point / center Jacobians here and avoid $\partial\vec{q}/\partial\vec{v}$ entirely in §7.

Unit vector. For $\hat{u} = \vec{u}/|\vec{u}|$, $\partial\hat{u}^\alpha/\partial u^\beta = |\vec{u}|^{-1}\Pi_u^{\alpha\beta}$ with

$$\Pi_u^{\alpha\beta} = \delta^{\alpha\beta} - \hat{u}^\alpha\hat{u}^\beta \quad (6.1)$$

the projection off $\hat{u}$. Each edge unit is a normalized vertex difference, so

$$\frac{\partial\hat{e}_j^\alpha}{\partial v_k^\beta} = \frac{1}{|\vec{e}_j|}\Pi_{e_j}^{\alpha\beta}(\delta_{z(j)k} - \delta_{jk}), \quad (6.2)$$

and (block above, twice, with $|\hat{e}_k - \hat{e}_{z'(k)}| = 2\cos(\theta_k/2)$)

$$\frac{\partial\hat{b}_k^\alpha}{\partial v_m^\beta} = \frac{1}{|\hat{e}_k - \hat{e}_{z'(k)}|}\Pi_{b_k}^{\alpha\gamma}\left(\frac{\partial\hat{e}_k^\gamma}{\partial v_m^\beta} - \frac{\partial\hat{e}_{z'(k)}^\gamma}{\partial v_m^\beta}\right). \quad (6.3)$$

Points. Product-rule each point ($\otimes$ the outer product, $c = \rho/\sin(\theta_k/2)$, $\partial t_k/\partial\vec{v}_m = (dt_k/d\theta_k)\partial\theta_k/\partial\vec{v}_m$ and likewise for c , with $\partial\theta_k/\partial\vec{v}_m$ from `cornerAngleGradients`):

$$\frac{\partial\vec{a}_k^+}{\partial\vec{v}_m} = \delta_{mk}I + \hat{e}_k \otimes \frac{\partial t_k}{\partial\vec{v}_m} + t_k \frac{\partial\hat{e}_k}{\partial\vec{v}_m}, \quad (6.4)$$

$$\frac{\partial\vec{a}_k^-}{\partial\vec{v}_m} = \delta_{mk}I - \hat{e}_{z'(k)} \otimes \frac{\partial t_k}{\partial\vec{v}_m} - t_k \frac{\partial\hat{e}_{z'(k)}}{\partial\vec{v}_m}, \quad (6.5)$$

$$\frac{\partial\vec{z}_k}{\partial\vec{v}_m} = \delta_{mk}I + \hat{b}_k \otimes \frac{\partial c}{\partial\vec{v}_m} + c \frac{\partial\hat{b}_k}{\partial\vec{v}_m}, \quad (6.6)$$

with the two scalar derivatives (the $\frac{1}{2}$ is $d(\theta_k/2)/d\theta_k$)

$$\frac{dt_k}{d\theta_k} = -\frac{\rho}{2\sin^2(\theta_k/2)}, \quad (6.7)$$

$$\frac{dc}{d\theta_k} = -\frac{\rho \cos(\theta_k/2)}{2\sin^2(\theta_k/2)} = -\frac{\rho}{2\sin(\theta_k/2)\tan(\theta_k/2)}. \quad (6.8)$$

7 Assembling the force

We avoid $\partial\vec{q}/\partial\vec{v}$ by differentiating the area through its boundary. With $\vec{a} \times \vec{b} = \varepsilon_{\alpha\beta} a^\alpha b^\beta$, vary $A_\cap = \frac{1}{2} \oint X \times dX$ and integrate by parts; the closed-loop boundary term vanishes, leaving

$$\delta A_\cap = \oint_{\partial(A\cap B)} \delta\vec{X} \times d\vec{X}. \quad (7.1)$$

The boundary point $\vec{X}$ is a function of the vertices, so a vertex motion $\{\delta\vec{v}_j\}$ gives $\delta\vec{X} = \sum_j (\partial\vec{X}/\partial\vec{v}_j) \delta\vec{v}_j$. By linearity, $\delta A_\cap = \sum_j [\oint (\partial\vec{X}/\partial\vec{v}_j) \times d\vec{X}] \delta\vec{v}_j$, and the coefficient of $\delta\vec{v}_j$ is the gradient,

$$\frac{\partial A_\cap}{\partial\vec{v}_j} = \oint_{\partial(A\cap B)} \frac{\partial\vec{X}}{\partial\vec{v}_j} \times d\vec{X}. \quad (7.2)$$

On a run lying on a feature, $\partial\vec{X}/\partial\vec{v}_j$ is the velocity of a point pinned to that feature at fixed parameter – the §6 Jacobians $\delta\vec{a}^\pm, \delta\vec{z}$. A run's endpoints are intersection points; moving one slides it tangentially along the boundary, which leaves the integral unchanged, so we keep the current endpoints and never form $d\vec{q}$. Evaluate $\int_{\text{piece}} \delta\vec{X} \times d\vec{X}$ per kept piece $\vec{P} \rightarrow \vec{Q}$ ($\vec{d} = \vec{Q} - \vec{P}$).

Edge piece (E_m). A point at edge fraction s is $\vec{X}(s) = (1-s)\vec{a}_m^+ + s\vec{a}_{z(m)}^-$, so $\delta\vec{X}(s) = (1-s)\delta\vec{a}_m^+ + s\delta\vec{a}_{z(m)}^-$ is linear in s . Reparametrize by $\lambda \in [0, 1]$, $\vec{X} = \vec{P} + \lambda\vec{d}$, $d\vec{X} = \vec{d}d\lambda$; bilinearity of $\times$ gives $\int_0^1 \delta\vec{X} \times \vec{d}d\lambda = \frac{1}{2}(\delta\vec{X}_P + \delta\vec{X}_Q) \times \vec{d}$, with $\frac{1}{2}(\delta\vec{X}_P + \delta\vec{X}_Q) = (1-\bar{s})\delta\vec{a}_m^+ + \bar{s}\delta\vec{a}_{z(m)}^-$, $\bar{s} = \frac{1}{2}(s_P + s_Q)$:

$$\delta A_{\text{edge},m} = [(1-\bar{s})\delta\vec{a}_m^+ + \bar{s}\delta\vec{a}_{z(m)}^-] \times \vec{d}. \quad (7.3)$$

Arc piece (C_m). A point at angle ϕ is $\vec{X}(\phi) = \vec{z}_m + \rho(\cos\phi, \sin\phi)$; the circle translates rigidly with $\vec{z}_m$, so $\delta\vec{X} = \delta\vec{z}_m$ is constant along the arc and $\int \delta\vec{X} \times d\vec{X} = \delta\vec{z}_m \times \int d\vec{X} = \delta\vec{z}_m \times \vec{d}$ ($\int d\vec{X} = \vec{d}$, the chord, path-independent):

$$\delta A_{\text{arc},m} = \delta\vec{z}_m \times \vec{d}. \quad (7.4)$$

Assembly. Write the kept piece's feature point as $\vec{f}_m$ (the bracket above on an edge, $\vec{z}_m$ on an arc) with chord $\vec{d}_m = \vec{Q} - \vec{P}$. Replacing $\delta\vec{X}$ by the Jacobian column $\partial\vec{f}_m/\partial\vec{v}_j$ in (7.2),

$$\frac{\partial A_\cap}{\partial v_j^\gamma} = \sum_{\text{features } m} \frac{\partial f_m^\alpha}{\partial v_j^\gamma} \varepsilon_{\alpha\beta} d_m^\beta, \quad (7.5)$$

$$\frac{\partial \vec{f}_m}{\partial \vec{v}_j} = \begin{cases} (1-\bar{s}_m) \frac{\partial \vec{a}_m^+}{\partial \vec{v}_j} + \bar{s}_m \frac{\partial \vec{a}_{z(m)}^-}{\partial \vec{v}_j} & \text{edge} \\ \frac{\partial \vec{z}_m}{\partial \vec{v}_j} & \text{arc,} \end{cases} \quad (7.6)$$

summed over the kept feature pieces of every overlapping pair. The force is $\vec{F}_j = -K_o \partial A_{\cap} / \partial \vec{v}_j$ (`overlapForce.py`). FD-checked to machine precision against $\partial(\text{overlapAreas}) / \partial \vec{v}$, with the net force summing to $\vec{0}$ (translation invariance) to $\sim 10^{-16}$.

8 The energy

The full potential is a sum of terms on the rounded polygon, $U = U_{\text{overlap}} + U_{\text{self}} + U_{\text{adh}} + U_A + U_P$; only the overlap term is active (the rest have coefficient 0).

$$U_{\text{overlap}} = K_o \sum_{i < j} |A_i \cap A_j| \quad (\text{active; force in §7}) \quad (8.1)$$

$$U_{\text{adh}} = -\frac{K_{\text{adh}}}{2} \sum_{\text{pairs}} \left(\frac{2\ell}{P_i^t + P_j^t} \right)^2 \quad (\text{off}), \quad (8.2)$$

$$U_A = \frac{K_A}{2} \sum_i (A_i - A_i^t)^2, \quad U_P = \frac{K_P}{2} \sum_i (P_i - P_i^t)^2 \quad (\text{off}), \quad (8.3)$$

with global stiffnesses $K_o, K_{\text{adh}}, K_A, K_P$ and per-polygon targets A_i^t, P_i^t . Self-repulsion (intra-polygon corner-circle avoidance) is also off. The area / perimeter springs need only their existing gradients (`roundedArea/PerimeterGradient`) switched on; adhesion and self-repulsion need their own derivations.